

Resolving the Dark Matter-IMF Degeneracy in JWST-ER1

Resolving the Dark Matter-IMF Degeneracy in Massive Quiescent Galaxies at $z \sim 2$ with Spatial...

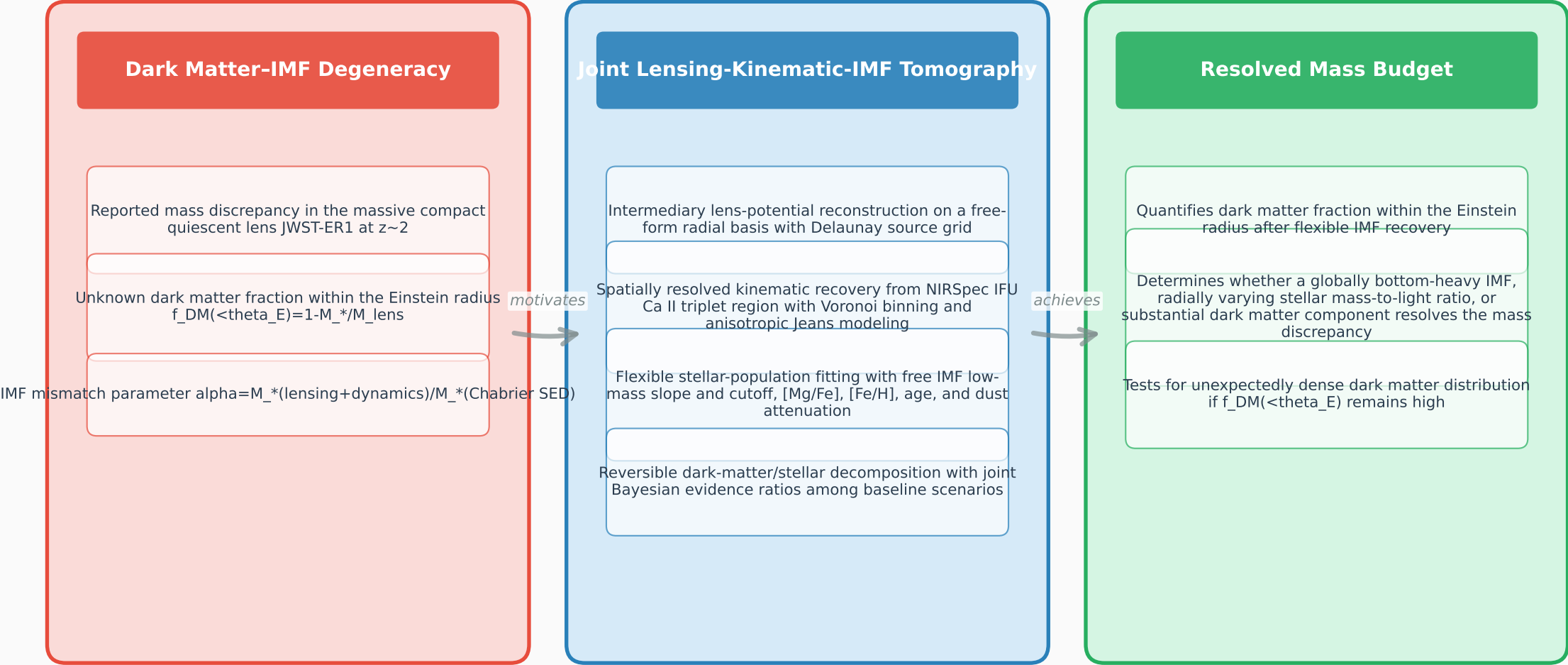


Figure 1: Overview of the JWST-ER1 joint strong-lensing, NIRSpec IFU kinematic, and spectro-photometric IMF tomography framework, in which the dark matter-IMF degeneracy motivates a free-form lens-potential reconstruction and reversible dark-matter/stellar decomposition to quantify the dark matter fraction within the Einstein radius and compare evidence ratios among Chabrier, bottom-heavy, and radially varying IMF scenarios.